



Diesel Generator Set

mtu 20V4000 DS2750

380V – 11 kV/50 Hz/standby power/NEA (ORDE) optimized
20V4000G64F/water charge air cooling



Optional equipment and finishing shown. Standard may vary.

Product highlights

Benefits

- Low fuel consumption
- Optimized system integration ability
- High reliability
- High availability of power
- Long maintenance intervals

Support

- Global product support offered

Standards

- Engine-generator set is designed and manufactured in facilities certified to standards ISO 2008:9001 and ISO 2004:14001
- Generator set complies to ISO 8528
- Generator meets NEMA MG1, BS 5000, ISO, DIN EN and IEC standards
- NFPA 110

Power rating

- System ratings: 2660 kVA - 2900 kVA
- Accepts rated load in one step per NFPA 110*
- Generator set complies to G3 according to ISO 8528-5
- Generator set exceeds load steps according to ISO 8528-5*

Performance assurance certification (PAC)

- Engine-generator set tested to ISO 8528-5 for transient response
- 85% load factor
- Verified product design, quality and performance integrity
- All engine systems are prototype and factory tested

Complete range of accessories available

- Control panel
- Power panel
- Circuit breaker/power distribution
- Fuel system
- Fuel connections with shut-off valve mounted to base frame
- Starting/charging system
- Exhaust system
- Mechanical and electrical driven radiators
- Medium and oversized voltage alternators

Emissions

- NEA (ORDE) optimized

Certifications

- CE certification option
- Unit certificate acc. to VDE-AR-N 4110

* Changes to the standard parameter sets (alternator-regulator and genset-controller) are necessary



A Rolls-Royce
solution

Application data ¹⁾

Engine			Liquid capacity (lubrication)	
Manufacturer		<i>mtu</i>	Total oil system capacity: l	300
Model		20V4000G64F	Engine jacket water capacity: l	205
Type		4-cycle	Intercooler coolant capacity: l	50
Arrangement		20V	Combustion air requirements	
Displacement: l		95.4	Combustion air volume: m³/s	2.7
Bore: mm		170	Max. air intake restriction: mbar	50
Stroke: mm		210	Cooling/radiator system	
Compression ratio		16.4	Coolant flow rate (HT circuit): m³/hr	80
Rated speed: rpm		1500	Coolant flow rate (LT circuit): m³/hr	32.5
Engine governor		ECU 9	Heat rejection to coolant: kW	980
Max power: kWm		2420	Heat radiated to charge air cooling: kW	410
Air cleaner		dry	Heat radiated to ambient: kW	105
Fuel system			Fan power for electr. radiator (40°C): kW	70
Maximum fuel lift: m		5	Exhaust system	
Total fuel flow: l/min		27	Exhaust gas temp. (after turbocharger): °C	540
Fuel consumption ²⁾			Exhaust gas volume: m³/s	7.1
At 100% of power rating:	l/hr	g/kwh	Maximum allowable back pressure: mbar	85
At 75% of power rating:	574.4	197	Minimum allowable back pressure: mbar	30
At 50% of power rating:	450.5	206		
	319.3	219		

Standard and optional features

System ratings (kW/kVA)

Generator model	Voltage	NEA (ORDE) optimized					
		without radiator			with mechanical radiator		
		kWel	kVA*	AMPS	kWel	kVA*	AMPS
Leroy Somer LSA52.3 UL16 (Low voltage Leroy Somer standard)	380 V	2240	2800	4254	2240	2800	4254
	400 V	2240	2800	4041	2240	2800	4041
	415 V	2240	2800	3895	2240	2800	3895
Leroy Somer LSA53.2 M12 (Low voltage Leroy Somer oversized)	380 V	2240	2800	4254	2240	2800	4254
	400 V	2240	2800	4041	2240	2800	4041
	415 V	2240	2800	3895	2240	2800	3895
Marathon 1020FDL7093 (Low voltage Marathon)	380 V	2272	2840	4315	2264	2830	4300
	400 V	2184	2730	3940	2184	2730	3940
	415 V	2128	2660	3701	2128	2660	3701
Marathon 1030FDL7094 (Low voltage Marathon oversized)	380 V	2272	2840	4315	2264	2830	4300
	400 V	2184	2730	3940	2184	2730	3940
	415 V	2128	2660	3701	2128	2660	3701
Marathon 1030FDH7101 (Medium volt. marathon)	11 kV	2312	2890	152	2248	2810	147
Leroy Somer LSA53.2 ZL12 (Medium volt. Leroy Somer)	11 kV	2320	2900	152	2240	2800	147

* cos phi = 0.8

1 All data refers only to the engine and is based on ISO standard conditions (25°C and 100m above sea level).
2 Values referenced are in accordance with ISO 3046-1. Conversion calculated with fuel density of 0.83 g/ml. All fuel consumption values refer to rated engine power.